

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Prime Factors, HCF & LCM

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PRIME FACTORS, HCF & LCM · CH1 · L3



ENG. ESLAM AHMED

The Map

This lesson collapses Prime Factors, HCF & LCM into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find HCF

TRIGGER *"work out", "find", a missing value*

03 Find LCM

TRIGGER *"work out", "find", a missing value*

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 25 website questions, 25 checked solutions, 3 local PDFs read.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5^{17} \times 5^2 = 5^{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q1

Use the index law

$$5^{17} \times 5^2 = 5^{19}$$

Find the LCM

$$800 = 8 \times 100$$

Finish: (a) 5^{19} , (b) $2^5 \times 5^2$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

- 1 (a) Write 517 times 52 as a single power of 5 (1) (b) Write 800 as a product of its prime factors. Show your working clearly. (2)

YOUR SOLUTION

02 Find HCF

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

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FORMULA BANK

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03 Find LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^8 \times 3^5 \times 11^4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q9

Find the LCM

$$A = 2^8 \times 3^5 \times 11^4$$

Find the LCM

$$\text{HCF} = 2^6 \times 3 \times 11^4$$

Finish: (a) $2^6 \times 3 \times 11^4$, (b) $2^9 \times 3^5 \times 11^8$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

9 A = 2 8 times 3 5 times 11 4 B = 2 6 times 3 times 11 8 (a) Find the highest common factor (HCF) of A and B. (2) (b) Find the lowest common multiple (LCM) of 2A and 3B. Give the LCM as a product of powers of its prime factors. (2)

YOUR SOLUTION

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q9

Find the LCM

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

Find the LCM

$$\text{LCM} = 2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

Finish: $2^4 \times 3^2 \times 5^4 \times 11 \times 13$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

9 A = 23 times 32 times 52 times 11 B = 24 times 3 times 54 times 13 Find the lowest common multiple (LCM) of A and B. Give your answer as a product of powers of prime numbers.

YOUR SOLUTION

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

2 Find the lowest common multiple (LCM) of 28 and 105

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^5 \times 5 \times 7^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1HR Q11

Find the LCM

$$A = 2^5 \times 5 \times 7^2$$

Find the LCM

$$5A = 2^5 \times 5^2 \times 7^2$$

Finish: (a) $2^4 \times 5^2 \times 7^2$, (b) $2^{16} \times 5^8 \times 7^{12}$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

11 A B 2 5 7 2 5 7 5 2 3 3 4 (a) Write down the highest common factor (HCF) of 5A and 2B Give your answer as a product of prime factors. (2) A B 2 5 7 2 5 7 5 2 3 3 4 (b) Work out the value of AB 2 Give your answer as a product of prime factors. (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write prime factor trees
"work out", "find", a missing value	Find HCF
"work out", "find", a missing value	Find LCM
"work out", "show", "hence", a familiar exam phrase	Use HCF/LCM in context
"work out", "find", a missing value	Find the LCM
"work out", "show", "hence", a familiar exam phrase	The HCF takes the smaller

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.